

IMAGE(S)

Fashion Choice as Indicators of Cultural Change

By Hans-Joachim Voth & David Yanagizawa-Drott

Presented by Jan Bauer, Yosef König, and Alexandros Sobczyk

Modes of self-expression reflect on contemporary culture

Can analysing these modes give insight how culture *changes* over time?



Key challenge in the visual analysis of cultural
change:

lack of high-frequency, granular data and
consistent measurements

Yearbook portraits
of US high school
seniors 1930-2010



Note: First row is from 1964, second row is from 1984.

**US post-war cultural
dominance**



**Distinctive visual
(fashion) changes
in culture**





Ronald Michaels



Tonya Morgan



Alicia Mortier



Jeremiah Mozal



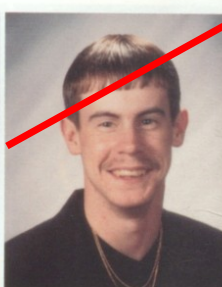
Elizabeth Murray



Kevin Muylaert



Nicole Nelson



Jeremy Nichols



Ryan Nyeholt



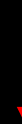
Aimee Owen



Wade Owens

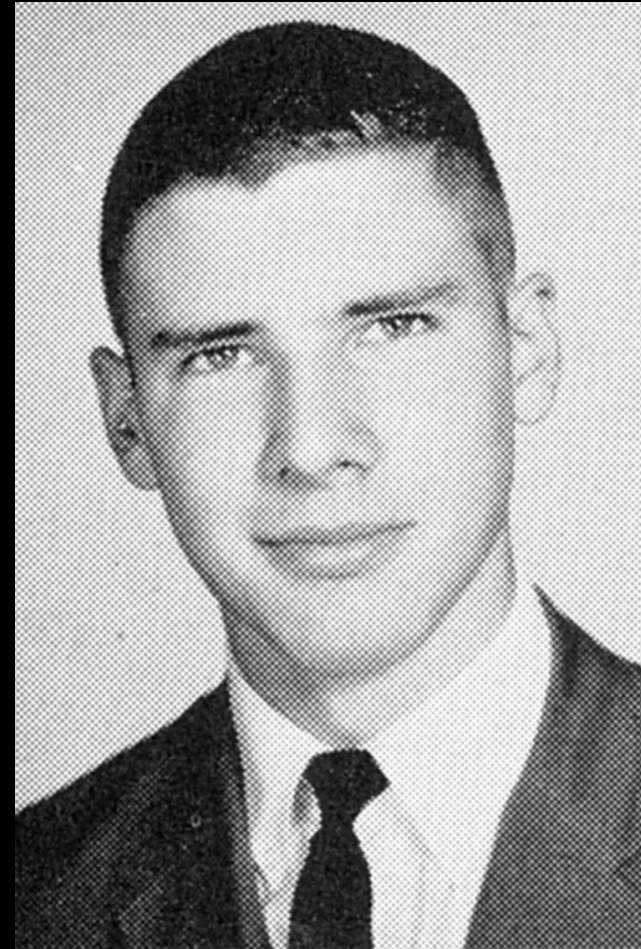


Iluminada Pagan



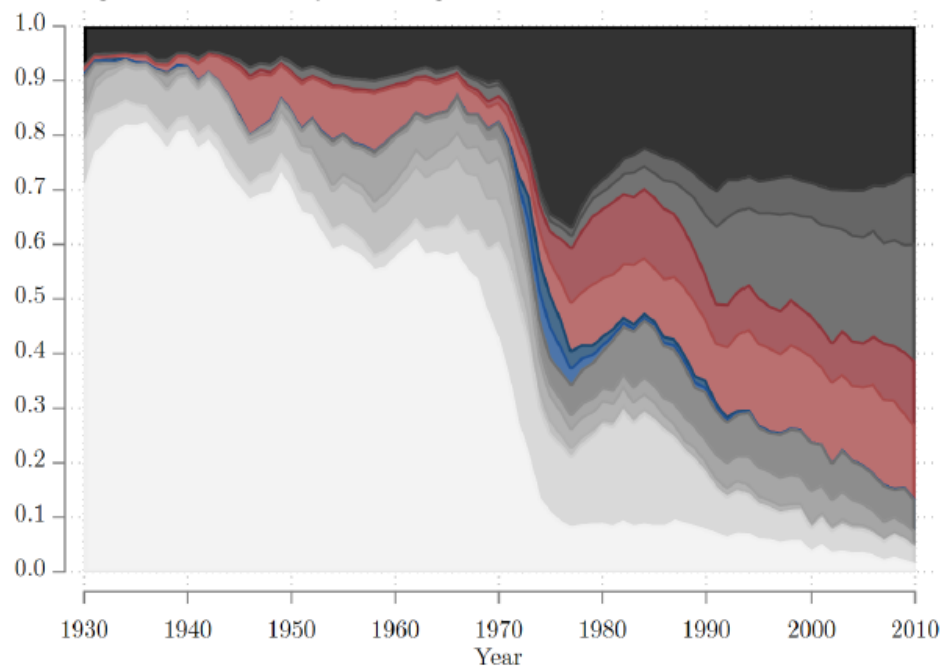
Classification of Style Markers:

- **Gender:** Male / Female (93%)
- **Hair:** Long / Short (91%)
- **Tie:** no tie / normal tie / bow tie (97%)
- **Clothing:** dress / suit / shirt without collar / shirt with collar (76%)
- **Glasses:** yes / no (99%)
- **Jewelery:** yes / no (91%)
- **Facial hair:** yes / no (70%)



Style Vector: HF = {male, short, normal tie, Suit, no, no, no}
(Style 221)

Figure 2: Males - Styles in High School Senior Yearbook Pictures.



999 Style	266 Style	265 Style	242 Style	241 Style
234 Style	233 Style	226 Style	225 Style	224 Style
223 Style	222 Style	221 Style		

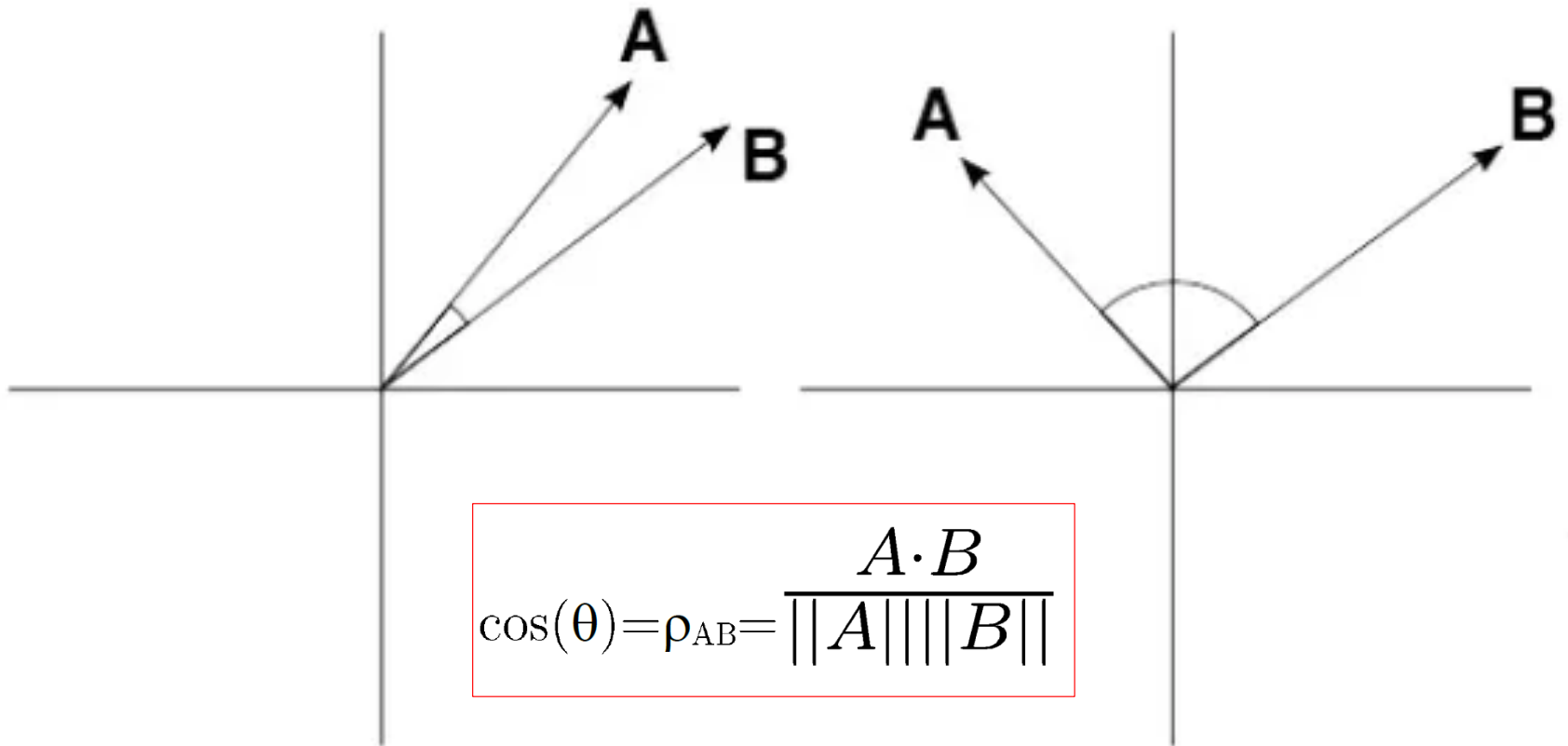
Style 221: suit/short hair/normal tie/no glasses/no facial hair/no jewelry
 Style 222: suit/short hair/normal tie/no glasses/yes facial hair/no jewelry
 Style 223: suit/short hair/normal tie/yes glasses/no facial hair/no jewelry
 Style 224: suit/short hair/normal tie/yes glasses/yes facial hair/no jewelry
 Style 225: suit/short hair/bow tie/no glasses/no facial hair/no jewelry
 Style 226: suit/short hair/bow tie/no glasses/yes facial hair/no jewelry
 Style 233: suit/long hair/normal tie/no glasses/no facial hair/no jewelry
 Style 234: suit/long hair/normal tie/no glasses/yes facial hair/no jewelry
 Style 241: shirt with collar/short hair/no tie/no glasses/no facial hair/no jewelry
 Style 242: shirt with collar/short hair/no tie/no glasses/yes facial hair/no jewelry
 Style 265: shirt without collar/short hair/no tie/no glasses/no facial hair/no jewelry
 Style 266: shirt without collar/short hair/no tie/no glasses/yes facial hair/no jewelry
 Style 999: Others

Style A: shirt / short hair / normal tie / no glasses / yes facial hair / yes jewellery

Style B: suit / short hair / normal tie / no glasses / yes facial hair / no jewellery

Similar

Unrelated



Individualism and persistence

- **Individualism:**

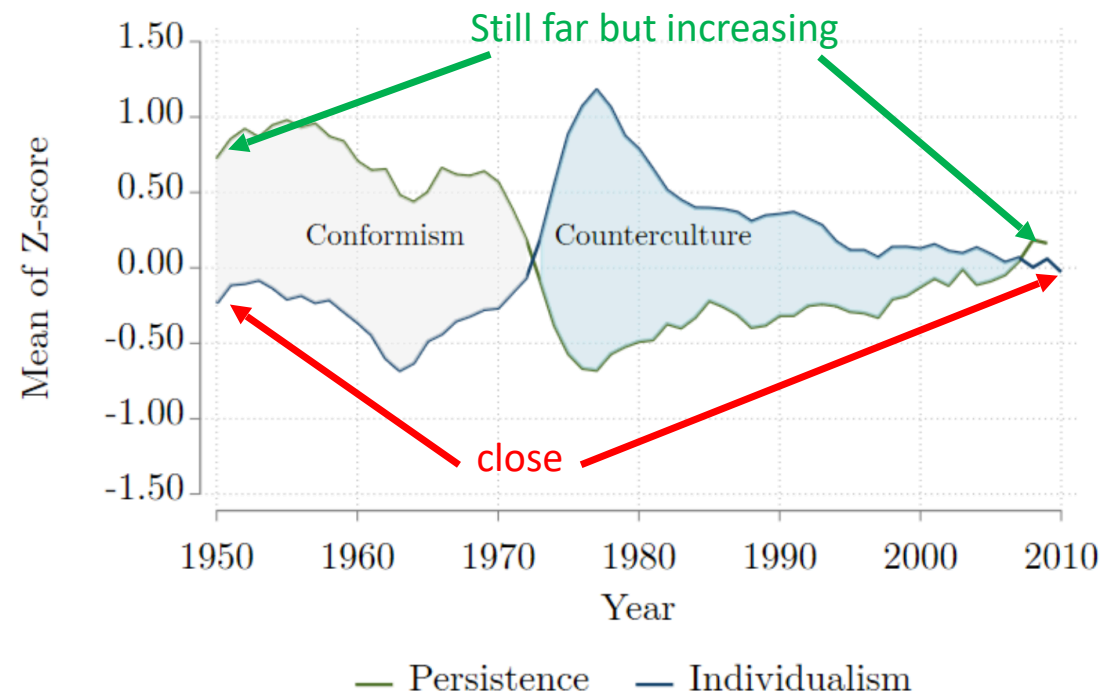
How **different** (“individual”) is a random student compared to their **classmates**

- **Persistence:**

How different are the students in the **same school** after **20 years**.

(Z-scored cosine similarity)

Figure 3: Individualism and Persistence in US High School Yearbook Images, 1950-2010.



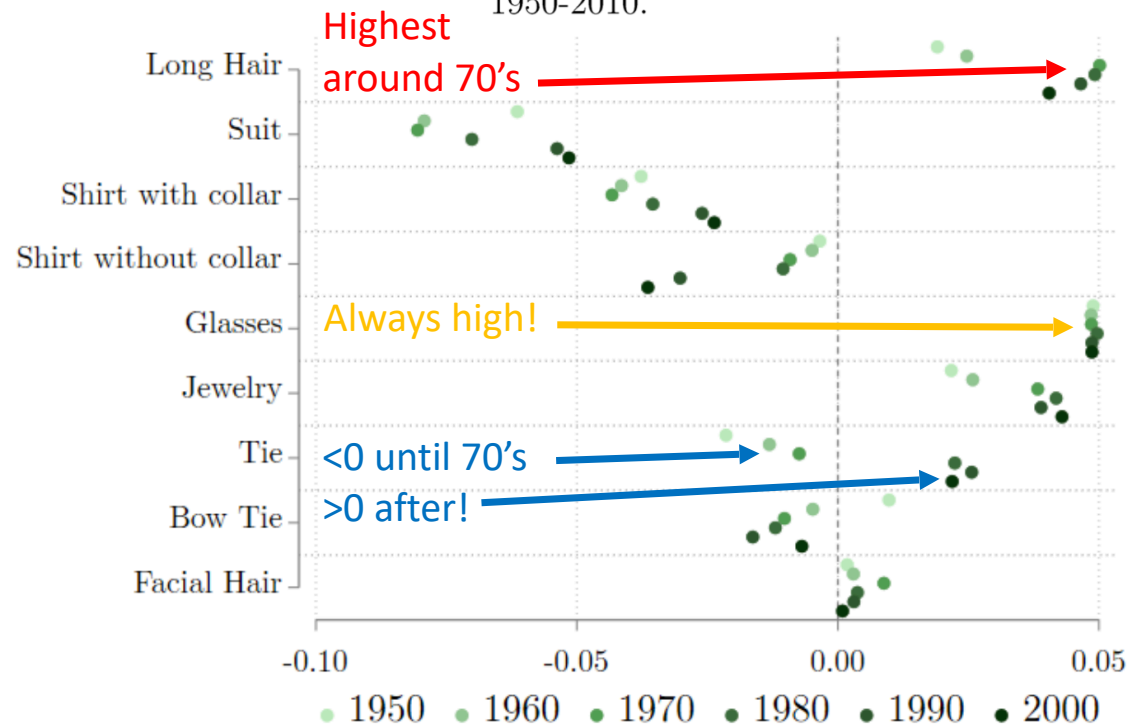
Note: All individuals across US, each compared with all individuals from same high school from their own year (individualism) and 20 years before (persistence). Cosine similarity is inverted for individualism, both are Z-scored.

What drove change?

- Quantify importance of factors based on LASSO regression model:

$$R(\beta) = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Figure 4: Drivers of Individualism in US High School Yearbook Images, 1950-2010.

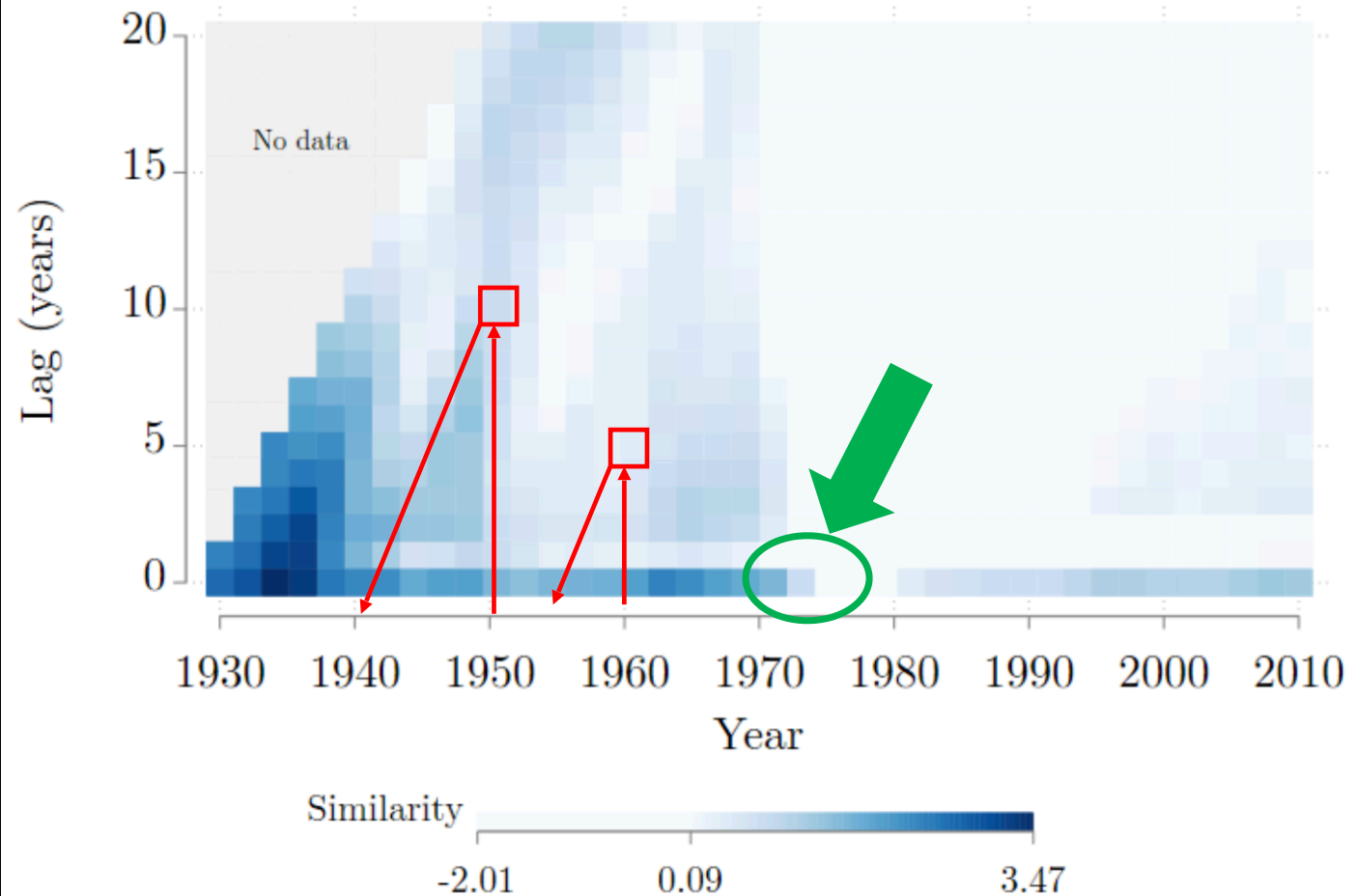


Note: Dots represent coefficient estimates from a Lasso regression of individualism score on style characteristics. Each dot represents a single coefficient for a style, derived from bi-variate regressions. Values greater zero indicate a positive contribution to students' individualism score.

Persistence analysis in-depth

- Compare against different “time windows”
- **Up to 1970:** good persistence up-to 15 years back
- **Early 70's:** sharp rise of counter-culture (almost zero persistence!)
- **After 80:** short-term persistence increases

Figure 5: Persistence over Time.



Innovative styles

- Which style is **innovative**?

- Should be **novel** and **influential**

- **Backward Similarity** of individual j (τ years):

$$BS_j^\tau = \sum_{i \in \mathcal{B}_{j,\tau}} \rho_{j,i}$$

- **Forward Similarity** of individual j (τ years):

$$FS_j^\tau = \sum_{i \in \mathcal{F}_{j,\tau}} \rho_{j,i}$$

- **Innovation score** of individual j : $q_j^\tau = \frac{FS_j^\tau}{BS_j^\tau}$

- BS **Low** + FS **High**:

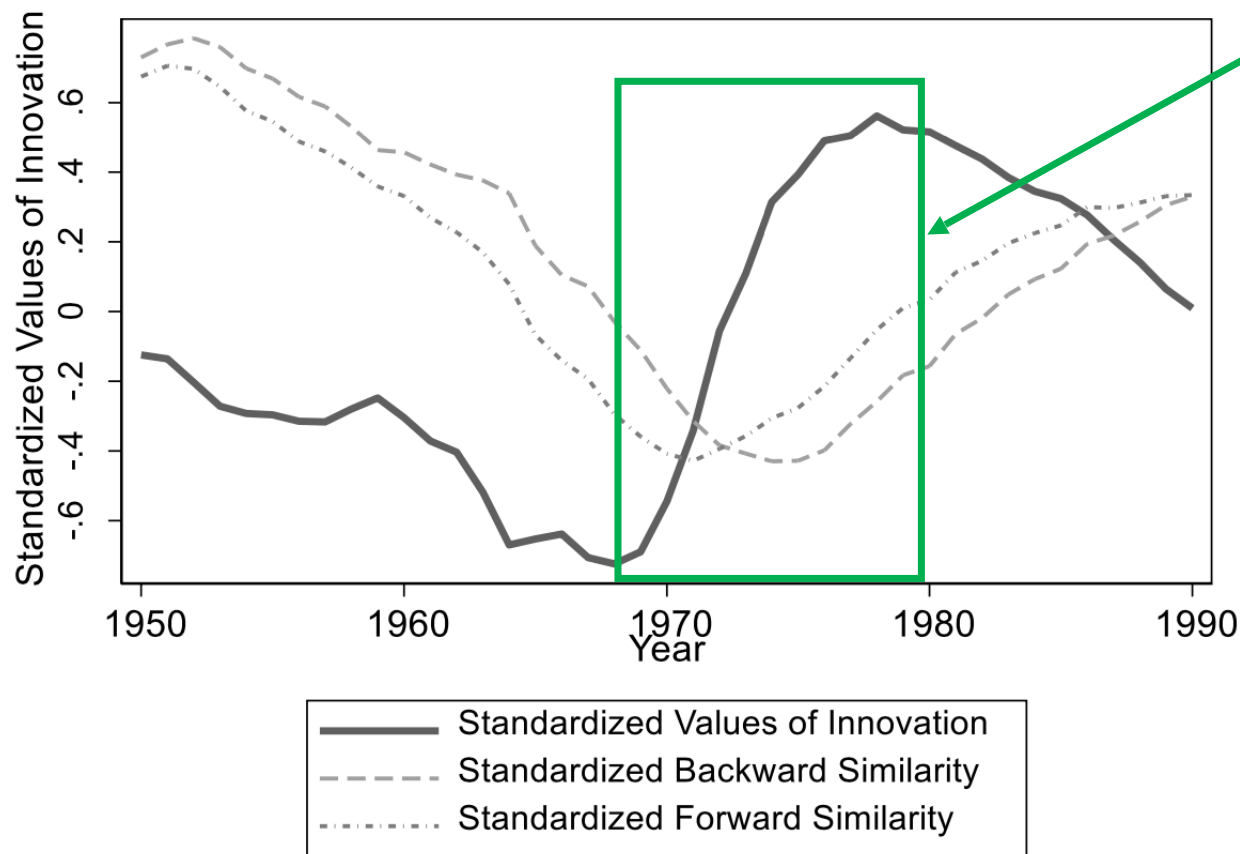
- **Innovative style**
- A style that was novel and influenced the others

- BS **Low** + FS **Low**

- **Failed innovation**
- A style that was novel but didn't catch up

Innovation: 5-year window

Figure 6: Influential Innovation in Style over Time, Decomposed into Forward and Backward Similarity



High innovation
in style in the
70's

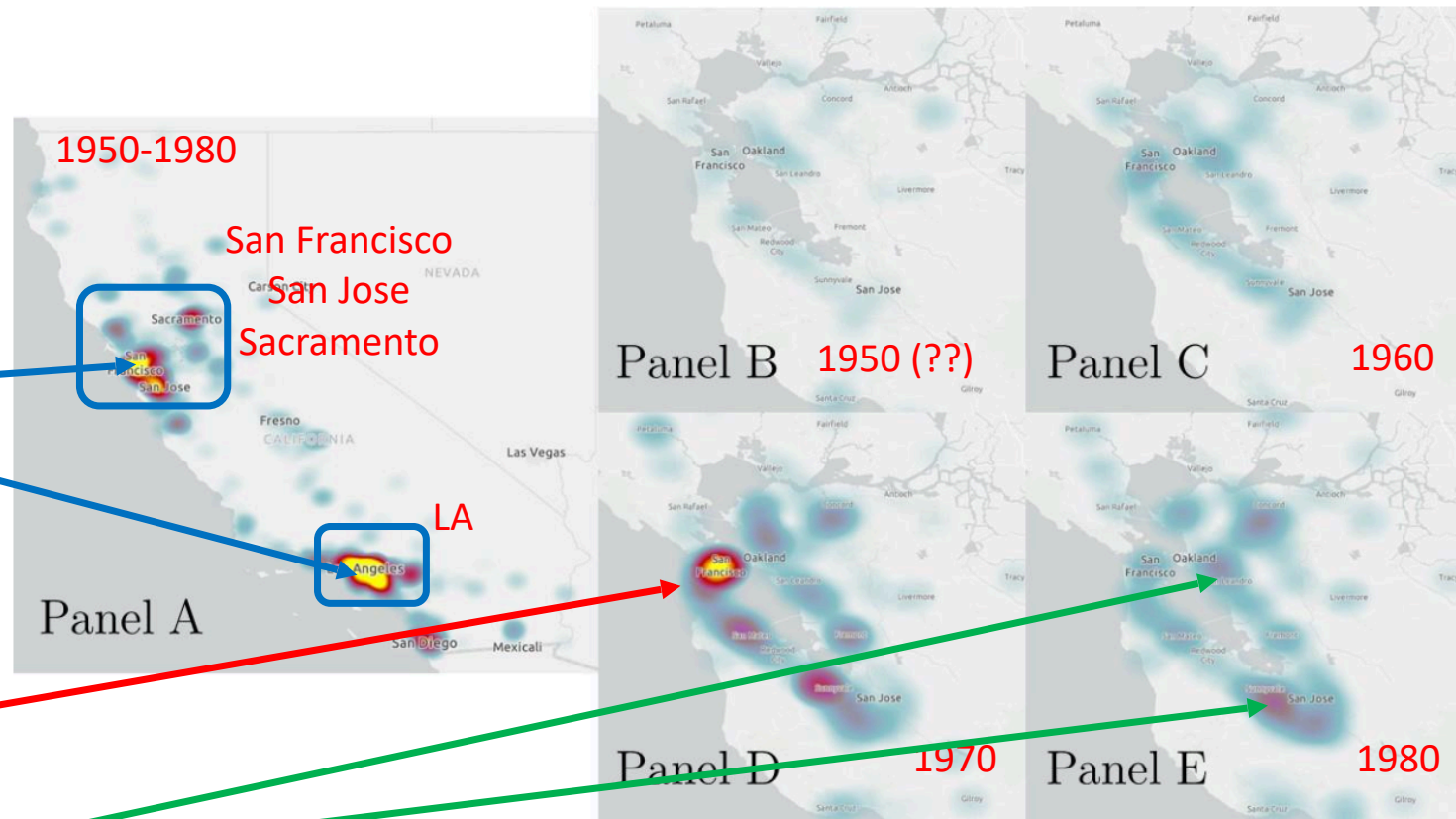
- Innovation of j:

$$q_j^\tau = \frac{FS_j^\tau}{BS_j^\tau}$$

Culturally leading areas

- Overall "centers of new style": SF+LA
- San Francisco dominates the 70's:
 - Hippie movement
 - Haight-Ashbury scene
- San Jose and Oakland dominate the 80's

Figure 7: Kelly et al. Measure of Innovation in California, 1960-1990

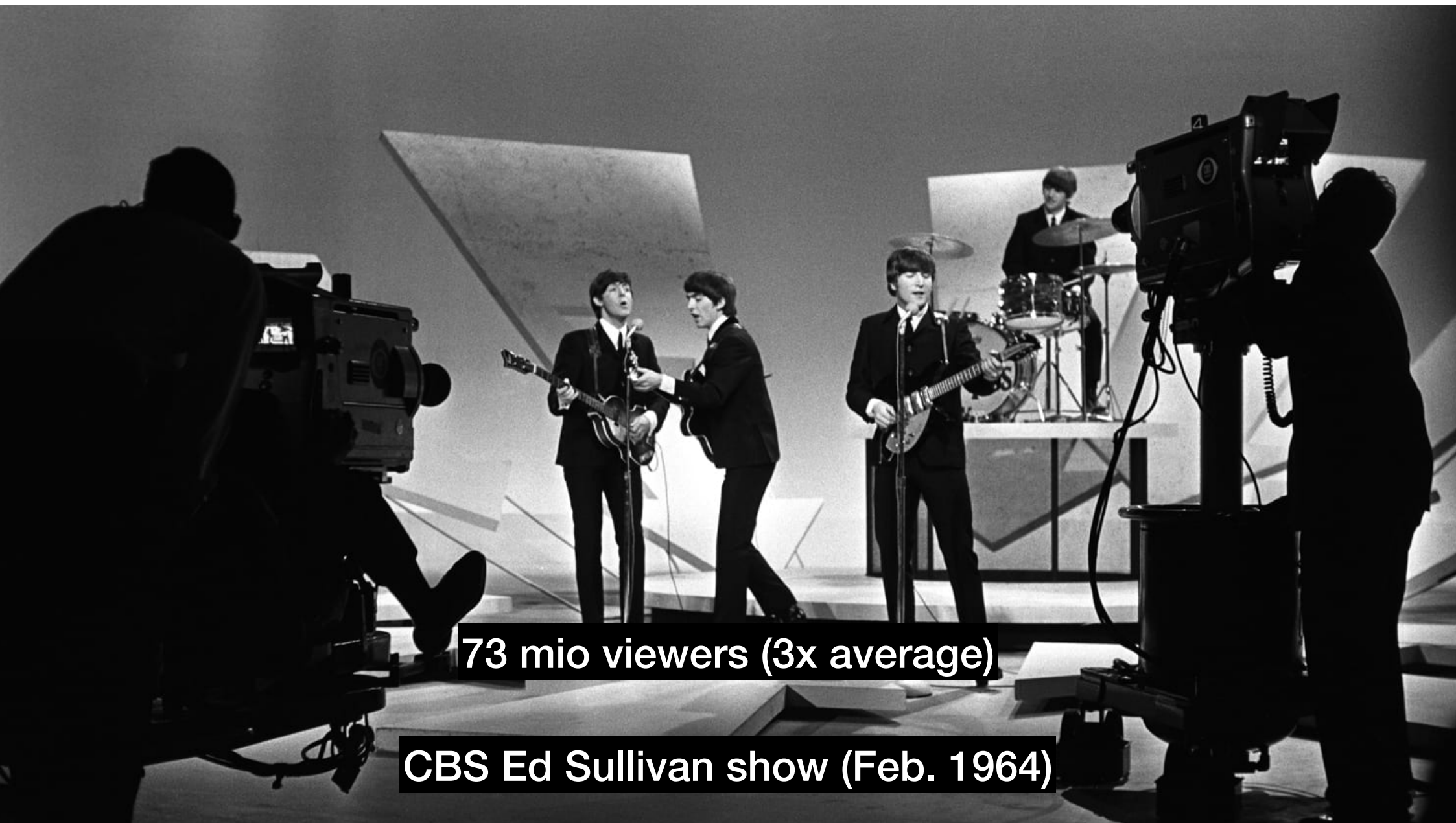


Note: For 25% of the images in California, we calculate the innovation index using a five year window forwards and backwards. We calculate city averages of innovation for each year from 1950 to 1980. Panel A shows a heat map of innovation according to the Kelly et al. measure for the period as a whole; Panel B is for 1950, C for 1960, D for 1970, and E for 1980.

**Can the data detect
rapid cultural change?**



Beatles 1963



73 mio viewers (3x average)

CBS Ed Sullivan show (Feb. 1964)



Fans on a roof at London Airport wave goodbye to the Beatles yesterday.

3d. Saturday, February 8, 1964

No. 18,704

YEAH! YEAH! U.S.A!



That old Beatlemania hits New York... a screaming girl tries to get nearer the Beatles.



Paul, Ringo, George and John answer questions at the Press conference.

February - 2.5 mio records sold

April - beatles singles occupy place 1-5 record sales

FATHER FLIES TO GET IRENE

PRINCESS Irene of Holland, whose romance has started a constitutional crisis, is going home today.

It was announced last night that her father, Prince Bernhard, would fly to Spain to pick her up.

And in Madrid, the Spanish capital, Irene's secretary stated early to-

'Engagement news soon'

Royal family is gathering for an important meeting when Irene arrives today. The announcement that Irene would be returning home today said that she had been spending several days in a "house of

From BARRIE HARDING New York, Friday

FIVE thousand screaming, chanting teenagers—most of them playing truant from school—gave the Beatles a fantastic welcome here today.

More than 100 extra police were on duty to control the crowd as the group's jet landed at the John F. Kennedy

5,000 scream 'welcome' to the Beatles

the world has gone mad. And a veteran airport employee said: "I see it—but I don't believe it."

There were screams and shouts as their guitars appeared on a luggage trolley. There were fresh squeals

crowd rivalled anything since General MacArthur returned from Korea.

The airport Press conference which followed the Beatles' arrival was chaos. Hundreds of reporters and photographers, plus seven T.V. cameras, had the room bursting at its seams.

Money

Part of the question-and-answer session between the porters and Beatles went like this:

"Will you sing something?"
John Lennon: "No!"
"Can you sing?"
"Not without money."
"How much money do

Maloney Fires No-Hitter at Cubs 1-0

Sec 4
Page 1

Written and Edited
MERIT YOUR *Confidence*

THE HOUSTON POST

SPORTS
FINAL

HOUSTON 1, TEXAS, FRIDAY, AUGUST 20, 1963

YEAR, NUMBER 21,314

FOUR SECTIONS

DIAL WANT ADS

TELEPHONE FA 4-3121

Published Every Morning by The Houston Post Company
Second Class Postage Paid at Houston, Texas

FIVE CENTS

mini 5
et for
urday

HOW SAILE
and KENNEDY

KENNEDY, Fla.
A blue sky over
S turned black
y— literally and
dy — and the
of the eight-day
sides was car-
I read for 8 AM
urday.

sting faded and
r muted away
nomen officials
nomy problems had
up too quickly.

AM Thomas then,
ini 5 light was
y E. E. O'Brien,
a director of the



MANIA

300 Beatle Fans Taken to Hospital

Singers
Protected
By Armor

By GEORGE CHRISTIAN

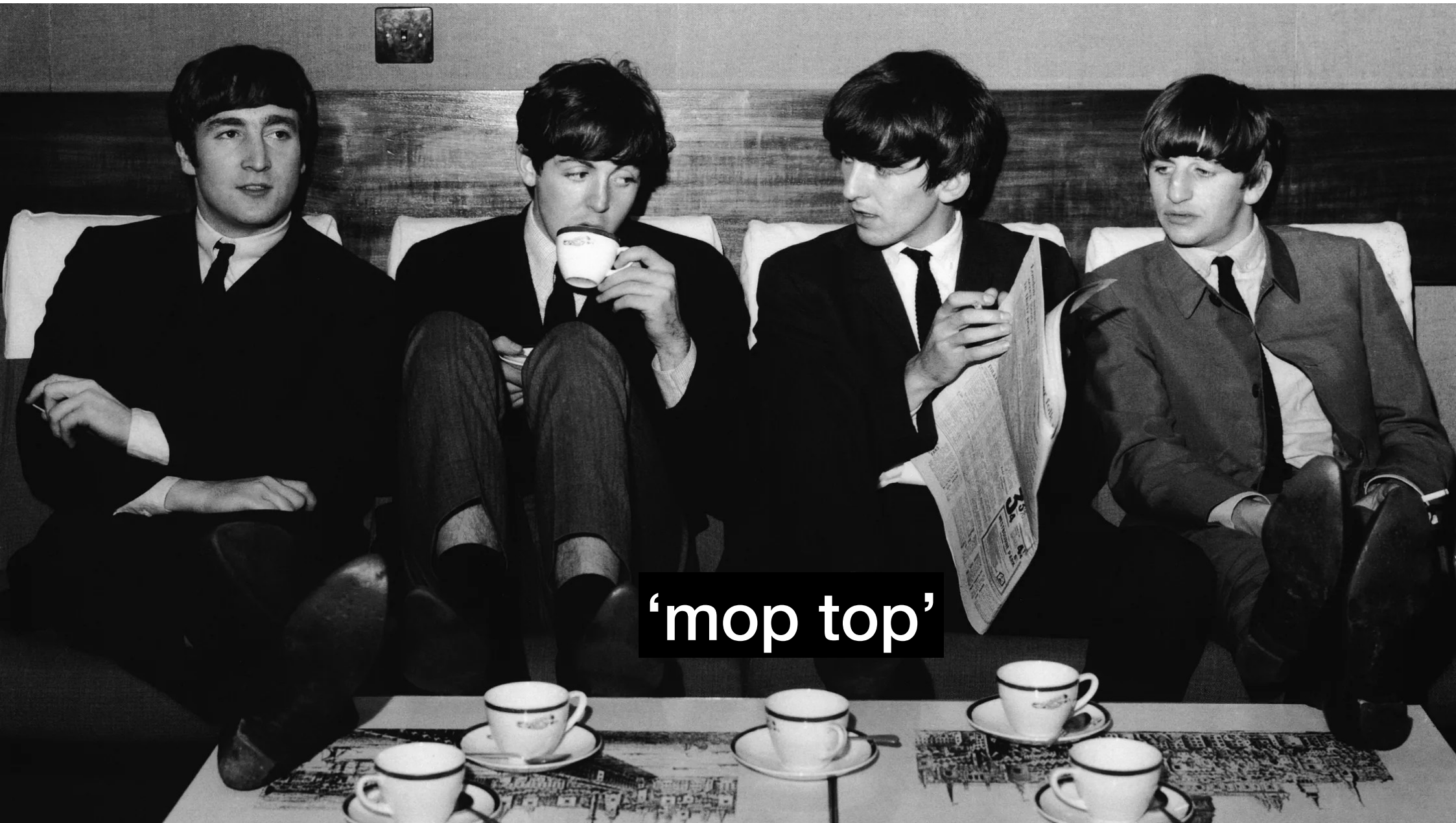
*Some Parents Secret Beatle
Fans, Psychiatrist Claims*

By LYNN BILLINGSLEY
Post Staff Writer

A lot of parents are secret
fans of the Beatles.

son, teen-agers and parents
have a contradiction with the
British Beatles. The singers
have an image of cleanliness.

And many psychiatrists
might think there is some psy-
chotic basis to all of this, but
I don't really know," he said.

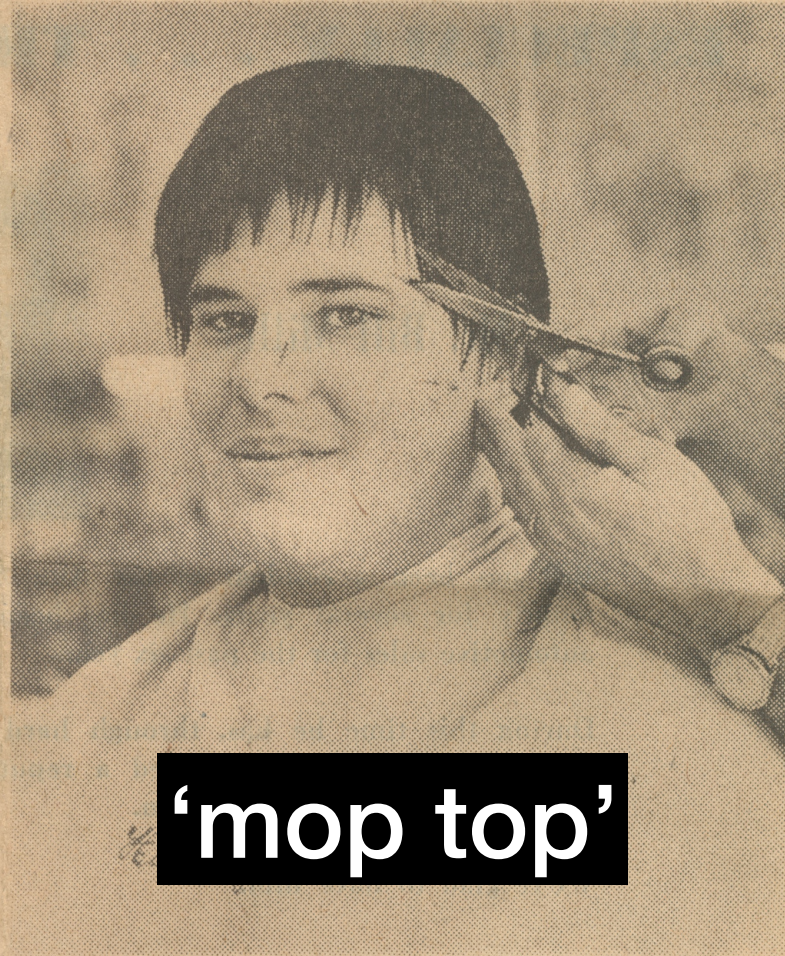


‘mop top’

Bugged By Beatles



Frederick F. Burrell, 21, of 821 Packard went right out Monday and bought a wig styled like hairdos of the members of the Beatles music combo, but found it needed a bit of trimming to suit him. The "haircut" was done at the Dascola Barbershop in Arborland. Why was Burrell, a sopho-



'mop top'

more at Eastern Michigan University in Ypsilanti, so eager to buy the wig? It seems that his girl friend is a Beatles fan and her enthusiasm was heightened by watching them on television Sunday night. "If I can't beat them, I'll join them," Burrell commented.

BOY FACES SUSPENSION —

Beatle Haircut Out At School

A 13-year-old boy who wants to be a musician or an actor is going to try Monday to get enrolled in school with a "sort of Beatle haircut."

He's going to try to enroll in the eighth grade at Orange Glen elementary school.

The boy — Donald Rogers of 1815 Summit Drive—hasn't been to a barbershop in six months but has let "almost anyone around" cut his hair on occasion. The result has been a rather neat, bowl-like haircut.

The boy is the son of Mrs. Nellie McHenry. His stepfather is George McHenry, a project engineer at General Dynamics.

Donald, informed that he would have to sport a more usual haircut, went to see Ron Jahelka, the new Orange Glen principal. Mrs. McHenry says that Donald told her he was informed by Jahelka that if he came to school Monday with the controversial haircut he would be taken home. If he returned to school still with the haircut, he would be suspended. Mrs. McHenry quotes him as saying he was told. And if he went back a third time, she related, he understood he would be expelled.



DONALD ROGERS,

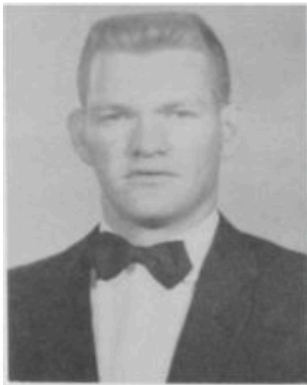
13-year-old Escondido boy, is shown here with his "sort of Beatle haircut" which he plans to wear to Orange Glen School on opening day Monday. He says he has been told he'll be returned home if he shows up with it at school.

★ ★ ★

Hair Style Classifier

p = mop top probability

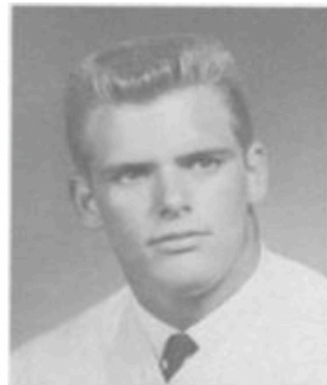
Accuracy 83%



$p = 0.01$



$p = 0.06$



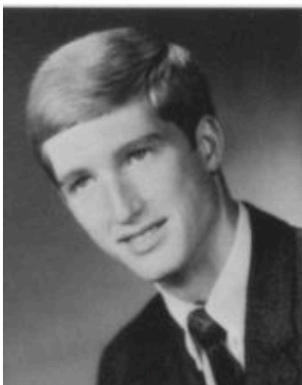
$p = 0.15$



$p = 0.25$



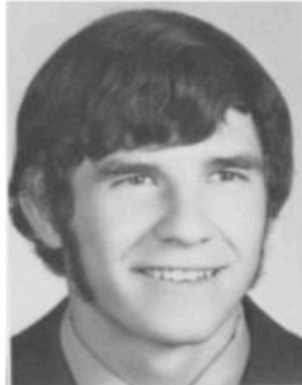
$p = 0.37$



$p = 0.52$



$p = 0.64$



$p = 0.75$

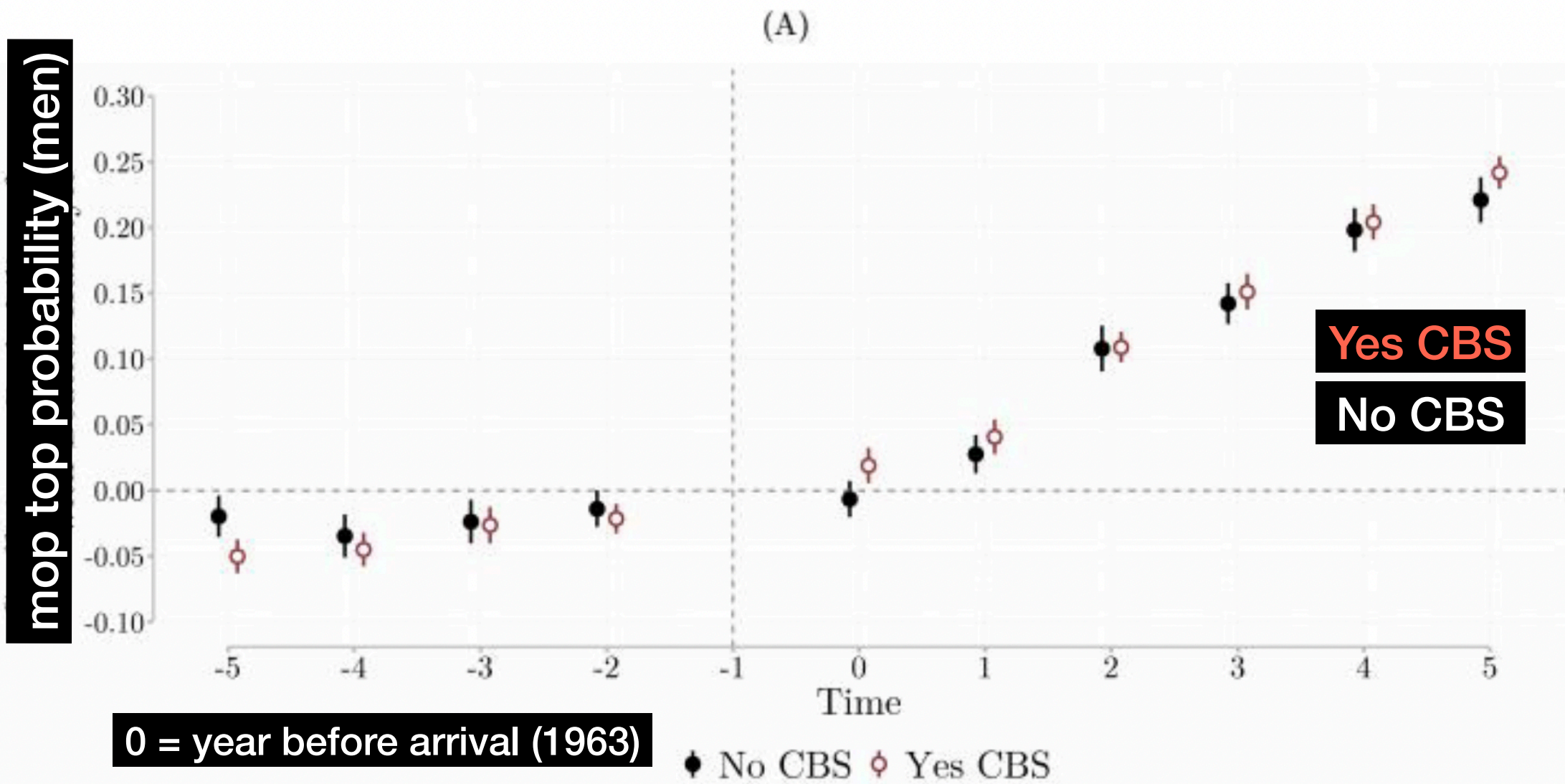


$p = 0.89$



$p = 0.95$

Figure 8: Beatles-style “Mop-Top” Hair in High School Senior Portraits

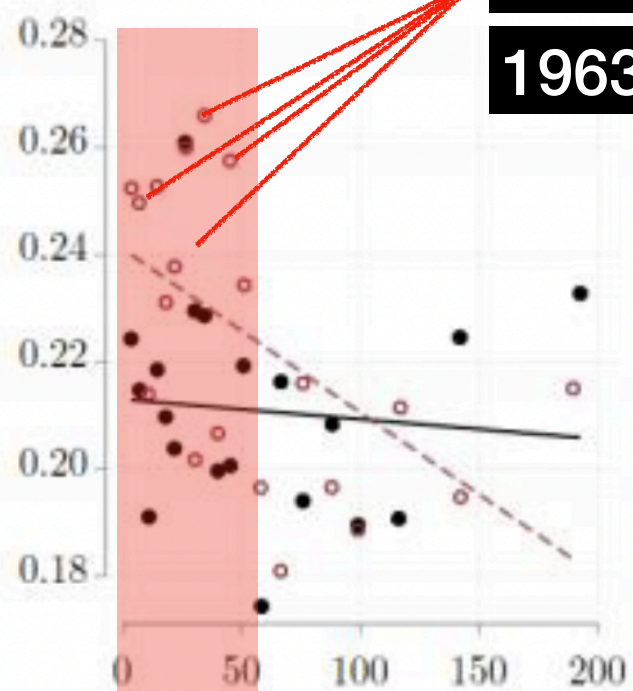


	No CBS	Yes CBS	Difference (No - Yes)
Total population, 1960	62897.705	172100.547	-109292.842**
Share of men, 1960 +	0.500	0.494	0.006**
Share non white people, 1960 +	0.092	0.089	0.004
Share of population (14 to 17) in schooling , 1960 -	0.869	0.877	-0.007
Share urban population, 1950	0.354	0.520	-0.166**
Unemployment Rate, 1950	0.042	0.041	0.001
Observations	319	318	

Descriptives are done at the county level. Table reports means and differences in demographic variables between CBS and No CBS group. We use Demographic Variables from Census 1950 and 1960 available at NHGIS. As the variable CBS/No CBS is constructed at high school level, there are some counties that have school with CBS signal and schools without CBS signal. Only counties without any school with CBS signal are considered as No CBS counties.

mop top probability (men)

(B)

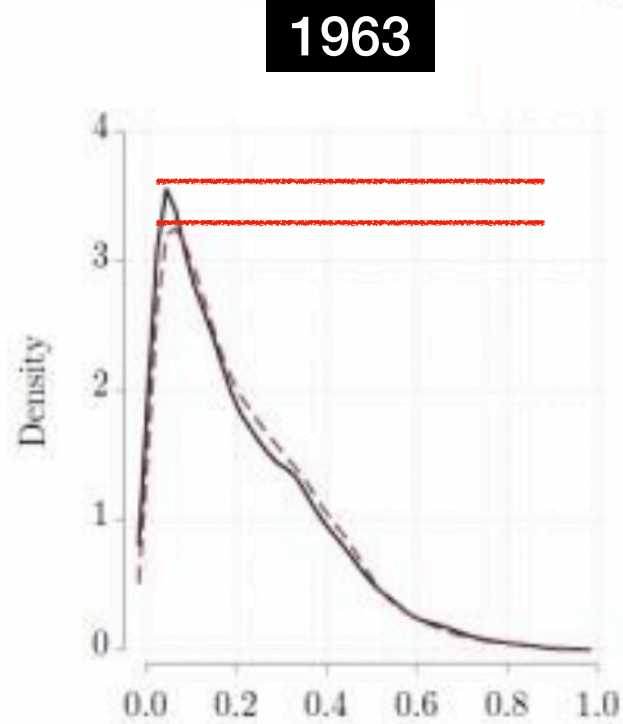


1964

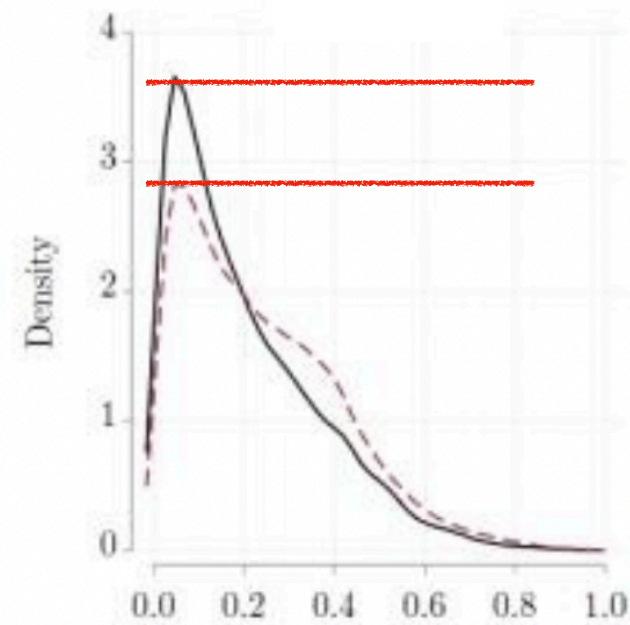
1963

Distance School - CBS Station

(C)



1964



Yes CBS

No CBS

—————

Discussion